HOMEWORK 4

Fourier Series

CPE 381

Instructor: Rahul Bhadani **Due: October 12, 2026, 11:59 PM**

125 points

You are allowed to use a generative model-based AI tool for your assignment. However, you must submit an accompanying reflection report detailing how you used the AI tool, the specific query you made, and how it improved your understanding of the subject. You are also required to submit screenshots of your conversation with any large language model (LLM) or equivalent conversational AI, clearly showing the prompts and your login avatar. Some conversational AIs provide a way to share a conversation link, and such a link is desirable for authenticity. Failure to do so may result in actions taken in compliance with the plagiarism policy.

Additionally, you must include your thoughts on how you would approach the assignment if such a tool were not available. Failure to provide a reflection report for every assignment where an AI tool is used may result in a penalty, and subsequent actions will be taken in line with the plagiarism policy.

Submission instruction:

Upload a .pdf on Canvas with the format `CPE381-LastFirst-HW-XX`. For example, if your name is Sam Wells, your file name should be `CPE381-WellsSam-HW-XX.pdf`. If there is a programming assignment, then you should include your source code along with your PDF files in a zip file `CPE381-LastFirst-HW-XX.zip`. Your submission must contain your name and UAH Charger ID or your UAH email address. Please number your pages as well.

Theory

1 Expand (10 points)

Find the Fourier coefficients for the given signal using definition of Complex Exponential Fourier Series: $x(t) = \cos(2\pi t) + \cos(3\pi t)$. Hint: find the fundamental period of $x(t)$ first. Show your work.

Answer

$$x(t) = \cos(2\pi t) + \cos(3\pi t)$$

$$\Omega_1 = 2\pi \Rightarrow T_1 = \frac{2\pi}{\Omega_1} = \frac{2\pi}{2\pi} = 1$$

$$\Omega_2 = 3\pi \Rightarrow T_2 = \frac{2\pi}{\Omega_2} = \frac{2\pi}{3\pi} = \frac{2}{3}$$

Fundamental time period of $x(t)$ is $L.C.M$ of $\left(1, \frac{2}{3}\right)$

$$= \frac{\text{LCM}(1,2)}{\text{GCD}(1,3)} = \frac{2}{1} = 2$$

Hence, the fundamental time period of $x(t)$ is

$$T_0 = 2 \text{ unit}$$

$$\Omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2} = \pi \text{ rad/unit}$$

$$x(t) = \cos(2\pi t) + \cos(3\pi t)$$

For exponential Fourier series:

$$X_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\Omega_0 t} dt$$

Using the fundamental period $T_0 = 2$ and $\Omega_0 = \pi$:

$$X_k = \frac{1}{2} \int_0^2 (\cos(2\pi t) + \cos(3\pi t)) e^{-jk\pi t} dt$$

Using Euler's identity $\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$:

$$= \frac{1}{2} \int_0^2 \left(\frac{e^{j2\pi t} + e^{-j2\pi t}}{2} + \frac{e^{j3\pi t} + e^{-j3\pi t}}{2} \right) e^{-jk\pi t} dt$$

$$= \frac{1}{4} \int_0^2 \left(e^{j2\pi t} + e^{-j2\pi t} + e^{j3\pi t} + e^{-j3\pi t} \right) e^{-jk\pi t} dt$$

$$= \frac{1}{4} \int_0^2 \left(e^{j(2\pi t - k\pi t)} + e^{j(-2\pi t - k\pi t)} + e^{j(3\pi t - k\pi t)} + e^{j(-3\pi t - k\pi t)} \right) dt$$

$$= \frac{1}{4} \int_0^2 \left(e^{j(2-k)\pi t} + e^{-j(2+k)\pi t} + e^{j(3-k)\pi t} + e^{-j(3+k)\pi t} \right) dt$$

$$X_k = \frac{1}{4}\left[\frac{e^{j(2-k)\pi t}}{j(2-k)\pi}\Big|_0^2 + \frac{e^{-j(2+k)\pi}}{-j(2+k)\pi t}\Big|_0^2 + \frac{e^{j(3-k)\pi t}}{j(3-k)\pi}\Big|_0^2 + \frac{e^{-j(3+k)\pi t}}{-j(3+k)\pi}\Big|_0^2\right]$$
$$= \frac{1}{4}\left[\frac{e^{j(2-k)2\pi}-1}{j(2-k)\pi} + \frac{e^{-j(2+k)2\pi}-1}{-j(2+k)\pi} + \frac{e^{j(3-k)2\pi}-1}{j(3-k)\pi} + \frac{e^{-j(3+k)2\pi}-1}{-j(3+k)\pi}\right]$$

Alternatively:

$$\Omega_0 = \frac{2\pi}{T_0} = \pi$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} c_n e^{jn\pi t}$$
$$\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$$
$$\cos(2\pi t) = \frac{e^{j2\pi t}+e^{-j2\pi t}}{2}$$
$$\cos(3\pi t) = \frac{e^{j3\pi t}+e^{-j3\pi t}}{2}$$
$$x(t) = \frac{1}{2}e^{j2\pi t} + \frac{1}{2}e^{-j2\pi t} + \frac{1}{2}e^{j3\pi t} + \frac{1}{2}e^{-j3\pi t}$$

Hence $X_k = \frac{1}{2}$ for $k = -3, -2, 2, -3$ and 0 for other k.

2 Half-wave Rectified Signal (15 Points)

Consider a portion of half-wave rectified signal $x(t)$ shown in Figure 1.
Its one period $x_1(t)$ can be written as

$$x_1(t) = \begin{cases} \sin(2\pi t), & 0 \le t \le 0.5 \\ 0, & 0.5 < t \le 1 \end{cases} \tag{1}$$

1. What is the fundamental period T_0 of the signal shown in the figure. **(2 Points)**

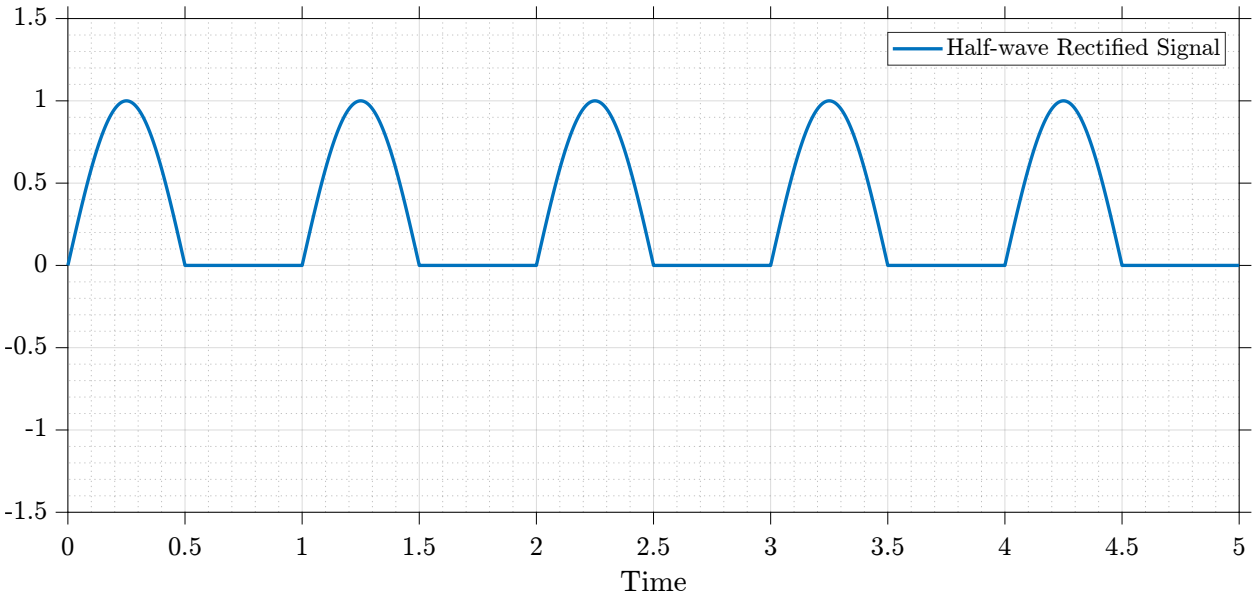


Figure 1: Half-wave Rectified Signal $x(t)$

2. Compute the complex exponential fourier series coefficient X_k using the integral definition of Fourier series coefficient. **(10 Points)**

3. Pay special attention to X_0 and X_1 and write their values separately. **(3 Points)**

Answer

For exponential Fourier series:

$$X_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\Omega_0 t} dt$$

The given fundamental period is $T_0 = 1$ unit. Hence, $\Omega_0 = 2\pi/1 = 2\pi$.
Then

$$
\begin{aligned}
X_k &= \int_0^1 x_1(t) e^{-jk2\pi t} dt \\
&= \int_0^{0.5} \sin(2\pi t) e^{-jk2\pi t} dt \\
&= \int_0^{0.5} \frac{e^{j2\pi t} - e^{-j2\pi t}}{2j} e^{-jk2\pi t} dt \\
&= \int_0^{0.5} \frac{e^{j2\pi t} - e^{-j2\pi t}}{2j} \\
&= \int_0^{0.5} \frac{e^{j2\pi(1-k)t} - e^{-j2\pi(1+k)t}}{2j} dt \\
&= \frac{1}{2j}\left[\frac{e^{j\pi(1-k)} - 1}{j2\pi(1-k)} - \frac{e^{-j\pi(1+k)} - 1}{-j2\pi(1+k)}\right]
\end{aligned}
\tag{2}
$$

We can get X_0 by putting $k = 0$ in the above equation, or alternatively, we put $k = 0$ in the integration, and we can obtain $X_0 = \frac{1}{\pi}$. We also need to deal with $k = 1$ and $k = -1$ separately, as directly putting $k = \pm 1$ will make $X_k \to \infty$.

$$
\begin{aligned}
X_0 &= \int_0^{0.5} \frac{e^{j2\pi t} - e^{-j2\pi t}}{2j} dt \\
&= \frac{1}{2j}\left[\int_0^{0.5} e^{j2\pi t} dt - \int_0^{0.5} e^{-j2\pi t} dt\right] \\
&= \frac{1}{2j}\left[\frac{e^{j2\pi t}}{j2\pi}\Big|_0^{0.5} - \frac{e^{-j2\pi t}}{-j2\pi}\Big|_0^{0.5}\right] \\
&= \frac{1}{2j}\left[\left(\frac{e^{j2\pi(0.5)}}{j2\pi} - \frac{e^{j2\pi(0)}}{j2\pi}\right) - \left(\frac{e^{-j2\pi(0.5)}}{-j2\pi} - \frac{e^{-j2\pi(0)}}{-j2\pi}\right)\right] \\
&= \frac{1}{2j}\left[\left(\frac{e^{j\pi} - 1}{j2\pi}\right) - \left(\frac{e^{-j\pi} - 1}{-j2\pi}\right)\right] \\
&= \frac{1}{2j}\left[\frac{e^{j\pi} - 1}{j2\pi} + \frac{e^{-j\pi} - 1}{j2\pi}\right] \\
&= \frac{1}{(2j)(j2\pi)}\left[e^{j\pi} + e^{-j\pi} - 2\right] \\
&= \frac{1}{-4\pi}\left[(2\cos(\pi)) - 2\right] \\
&= \frac{1}{-4\pi}\left[2(-1) - 2\right] \\
&= \frac{1}{-4\pi}\left[-2 - 2\right] \\
&= \frac{1}{-4\pi}(-4) \\
&= \frac{1}{\pi}
\end{aligned}
$$

$$
X_1 = \int_0^{0.5} \sin(2\pi t) e^{-j2\pi t} dt = \int_0^{0.5} \frac{1 - e^{-j4\pi t}}{2j} dt = \frac{1}{4j} - \frac{1}{8\pi} + \frac{1}{8\pi} = \frac{1}{4j}.
$$

Similarly, we can do for X_{-1}.

3 Graph Detective (20 points))

For the square wave given in Figure 4,

1. What is the Time period of the square wave? What is the Amplitude? **(5 points)**
2. Compute the Fourier series coefficients for the complex exponential Fourier series as well as trigonometric Fourier series representation using the integral method. Refer to page 255 of the textbook. Remember each term in the Fourier series represents a harmonic.

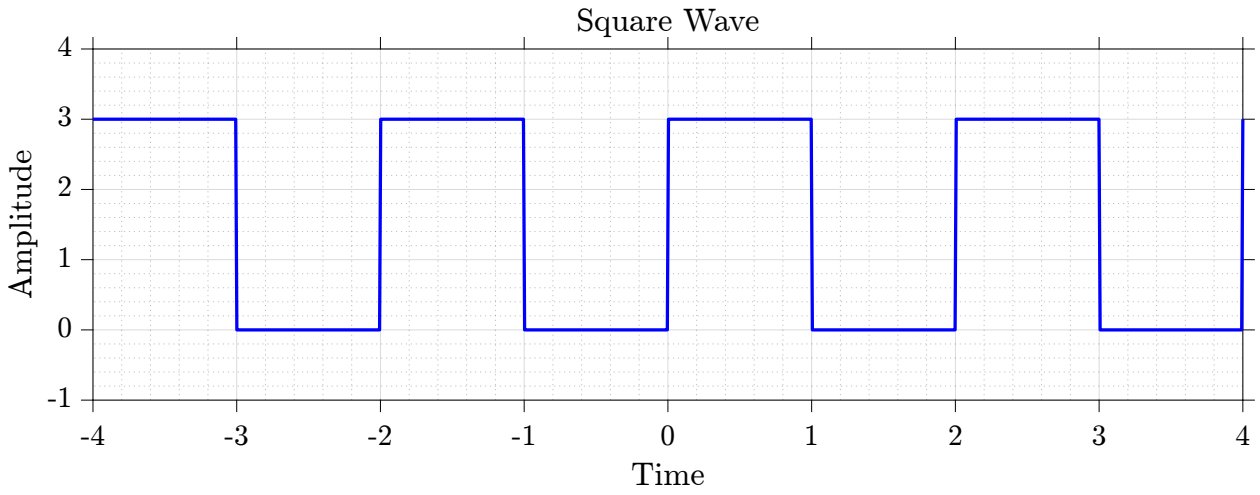


Figure 2: Square Wave $x(t)$

Write down the expression for X_k, c_k, and d_k. **(15 points)**.

Answer

1. The fundamental time period $T_0 = 2$ units.

$$\Omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2} = \pi$$

2.

$$\begin{aligned} X_k &= \frac{1}{2}\int_0^2 x(t)e^{-jk\pi t}dt \quad \text{for } t_0 = 0, T_0 = 2 \\ &= \frac{1}{2}\int_0^1 3 \cdot e^{-jk\pi t}dt \\ &= \frac{3}{2}\frac{e^{-jk\pi t}}{-jk\pi}\Big|_0^1 = \frac{3}{2}\left[\frac{e^{-jk\pi} - 1}{-jk\pi}\right] \\ &= \frac{3}{2}\left[\frac{(-1)^k - 1}{-jk\pi}\right] \\ &= \frac{3}{2}\left[\frac{1 - (-1)^k}{jk\pi}\right], \quad k \neq 0 \end{aligned} \tag{3}$$

$$X_0 = \frac{3}{2}\int_0^1 1dt = \frac{3}{2} \tag{4}$$

For trigonometric Fourier Series:

$$x(t) = c_0 + 2\sum_{k=1}^{\infty}[c_k \cos(k\Omega_0 t) + d_k \sin(k\Omega_0 t)] \tag{5}$$

And

$$c_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\cos(k\Omega_0 t)dt \tag{6}$$

Where $\Omega_0 = \pi$ and $T_0 = 2$.

$$\begin{aligned}
&= \frac{1}{2}\int_0^2 x(t)\cos(k\pi t)dt \\
&= \frac{1}{2}\int_0^1 3\cos(k\pi t)dt \\
&= \frac{3}{2}\left[\frac{\sin(k\pi t)}{k\pi}\right]_0^1 \\
&= \frac{3}{2}\frac{1}{k\pi}[\sin(k\pi\cdot 1) - \sin(k\pi\cdot 0)] \\
&= \frac{3}{2}\frac{1}{k\pi}[0-0] = 0
\end{aligned}$$

Thus, $c_k = 0$, for $k = 1, 2, \ldots$

$$\begin{aligned}
d_k &= \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\sin(k\Omega_0 t)dt \\
&= \frac{1}{2}\int_0^2 x(t)\sin(k\pi t)dt \\
&= \frac{1}{2}\int_0^1 3\sin(k\pi t)dt \\
&= -\frac{3}{2}\left[\frac{1}{k\pi}\cos(k\pi t)\right]_0^1 \\
&= -\frac{3}{2k\pi}[\cos(k\pi) - \cos(0)] \\
&= -\frac{3}{2k\pi}[(-1)^k - 1]
\end{aligned}$$

So,

$$d_k = \frac{3}{2k\pi}[1-(-1)^k] \tag{7}$$

So,

$d_k = \dfrac{3}{k\pi}$ for odd k.

For $k = 0$:

$$\begin{aligned} c_0 &= \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)dt \\ &= \frac{1}{2}\int_0^2 x(t)dt \\ &= \frac{1}{2}\int_0^1 3dt \\ &= \frac{1}{2}[3t]_0^1 \\ &= \frac{1}{2}[3(1) - 3(0)] = \frac{3}{2} \end{aligned}$$

Therefore,

$$x(t) = \frac{3}{2} + \sum_{k \text{ odd}}^{\infty} \frac{3}{k\pi}\sin(k\pi t) \tag{8}$$

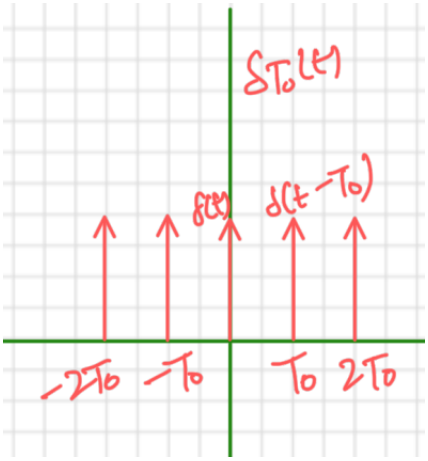
4 Fourier Series of a Pulse Training (10 Points)

Consider a periodic impulse train $\delta_{T_0}(t)$ as defined below:

$$\delta_{T_0}(t) = \sum_{-\infty}^{\infty} \delta(t - kT_0)$$

1. Sketch the impulse train. **(2 Points)**
2. Determine the complex exponential Fourier series of $\delta_{T_0}(t)$. **(3 Points)**
3. Determine the trigonometric Fourier series of $\delta_{T_0}(t)$.

Answer



$$\delta_{T_0}(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\Omega_0 t} \tag{9}$$

$$\Omega_0 = \frac{2\pi}{T_0} \tag{10}$$

$$X_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)e^{-jk\Omega_0 t}dt \quad = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta_{T_0}(t)e^{-jk\Omega_0 t}dt = \frac{1}{T_0} \quad (\text{Set } t_0 = -T_0/2) \tag{11}$$

$$\text{Hence } \delta_{T_0}(t) = \sum_{-\infty}^{\infty} \delta(t - kT_0) = \frac{1}{T_0}\sum_{K=-\infty}^{+\infty} e^{jk\Omega_0 t}\ \Omega_0 \quad = \frac{2\pi}{T_0} \tag{12}$$

$$\delta_{T_0}(t) = X_0 + 2\sum_{k=1}^{\infty}[c_k \cos(k\Omega_0 t) + d_k \sin(k\Omega_0 t)] \tag{13}$$

$$c_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\cos(k\Omega_0 t)dt = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t)\cos(k\Omega_0 t)dt \quad = \frac{1}{T_0} \tag{14}$$

$$d_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\sin(k\Omega_0 t)dt \quad = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t)\sin(k\Omega_0 t)dt = 0 \tag{15}$$

$$X_0 = \frac{1}{T_0} \quad (\text{From exponential Fourier Series expansion}) \tag{16}$$

$$\text{Thus } \delta_{T_0}(t) = \frac{1}{T_0} + \frac{2}{T_0}\sum_{k=1}^{\infty}\cos k\Omega_0 t \qquad \Omega_0 = \frac{2\pi}{T_0} \tag{17}$$

Practice

5 Harmonics in MATLAB (40 points)

1. Write a MATLAB script with the file name `harmonics_square_wave.m` to reproduce the square wave shown in Figure 4. If you make submission with any other file name, no points will be awarded. If you decided provide your implementation using functions, be sure put the all the functions at the end in your MATLAB script. **(10 points)**

2. From the previous question, you will obtain the formula for X_k. As you see other than the DC term X_0, k varies from $-\infty$ to ∞.

   Using your MATLAB program, approximate the square wave $x(t)$ using the first 11 terms from your trigonometric Fourier series which would be from $k = -5$ to $k = 5$. **(10 points)**.

3. Now add 21 terms (consider negative k and positive k equally), and then 51 terms (consider negative k and positive k equally). What do you observe? Explain. **(10 points)**.

4. Plot the magnitude spectrum, i.e. $|X_k|$ vs the index k **(5 points)**.

5. Plot the phase spectrum, i.e. $\angle X_k$ vs the index k **(5 points)**.

Answers:

Code also at https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/HW04_Harmonics.m

```
close all;
% Define the parameters
T = 2;  % Period of the square wave
N = 100;  % Number of Fourier series terms
t = linspace(-2*T, 2*T, 1000);  % Time array
Amplitude = 3.0;

% Define the square wave function
squarewave = @(t) Amplitude.*(mod(t, T) < T/2);

% Plot the results
f = figure;
f.Position = [517 687 1464 551];
plot(t, squarewave(t), 'b', 'LineWidth',2);
hold on;
xlabel('Time (s)');
title('Square Wave','Interpreter', 'latex');
%legend('Square Wave', 'Fourier Series Approximation');
set(gca, 'XColor', [0, 0, 0], 'YColor', [0, 0, 0], 'TickDir', 'out');
xaxis = get(gca, 'XAxis');
xaxis.TickLabelInterpreter = 'latex';
```

```
yaxis = get(gca, 'YAxis');
yaxis.TickLabelInterpreter = 'latex';
xlabel('Time','Interpreter', 'latex');
ylabel('Amplitude','Interpreter', 'latex');
set(gca, 'FontSize', 18);
ylim([-4,4])
grid on;
grid minor;
exportgraphics(f, '../figures/CPE381_FA24_HW04_Q5_Square_Wave.pdf');


Omega  = (2.*pi)./T;


k_end = 10;
x_t_10 = zeros(size(t));
for t_index = 1:length(t)
    for k = -k_end/2:1:k_end/2
      x_t_10(t_index) = x_t_10(t_index) + X_k(k).*exp(1j.*k.*Omega.*t(t_index));
    end
end

k_end = 20;
x_t_20 = zeros(size(t));
for t_index = 1:length(t)
    for k = -k_end/2:1:k_end/2
      x_t_20(t_index) = x_t_20(t_index) + X_k(k).*exp(1j.*k.*Omega.*t(t_index));
    end
end
k_end = 50;
x_t_50 = zeros(size(t));
for t_index = 1:length(t)
    for k = -k_end/2:1:k_end/2
      x_t_50(t_index) = x_t_50(t_index) + X_k(k).*exp(1j.*k.*Omega.*t(t_index));
    end
end


f = figure;
f.Position = [517 687 1464 551];
plot(t, squarewave(t), 'b', 'LineWidth',2, ...
    'DisplayName', 'Original Signal');
hold on;
plot(t, x_t_10, 'r', 'LineWidth',2, 'DisplayName','10 Harmonics');
```

```
plot(t, x_t_20, 'g', 'LineWidth',2, 'DisplayName','20 Harmonics');
plot(t, x_t_50, 'k', 'LineWidth',2, 'DisplayName','50 Harmonics');
hold on;
xlabel('Time (s)');
title('Square Wave','Interpreter', 'latex');
%legend('Square Wave', 'Fourier Series Approximation');
set(gca, 'XColor', [0, 0, 0], 'YColor', [0, 0, 0], 'TickDir', 'out');
xaxis = get(gca, 'XAxis');
xaxis.TickLabelInterpreter = 'latex';
yaxis = get(gca, 'YAxis');
yaxis.TickLabelInterpreter = 'latex';
xlabel('Time','Interpreter', 'latex');
ylabel('Amplitude','Interpreter', 'latex');
set(gca, 'FontSize', 18);
ylim([-4,4])
grid on;
legend;
grid minor;

exportgraphics(f, '../figures/CPE381_FA24_HW04_Q5_Harmonics.pdf');


% Magnitude Spectrum
k = -50:1:50;
Xk = X_k(k);

f = figure;
f.Position = [517 687 1464 551];
subplot(2,1,1)
stem(k, abs(Xk), 'b', 'LineWidth',2);
hold on;
xlabel('Time (s)');
title('Magnitude Spectrum','Interpreter', 'latex');
%legend('Square Wave', 'Fourier Series Approximation');
set(gca, 'XColor', [0, 0, 0], 'YColor', [0, 0, 0], 'TickDir', 'out');
xaxis = get(gca, 'XAxis');
xaxis.TickLabelInterpreter = 'latex';
yaxis = get(gca, 'YAxis');
yaxis.TickLabelInterpreter = 'latex';
xlabel('k','Interpreter', 'latex');
ylabel('$|X_k|$','Interpreter', 'latex');
set(gca, 'FontSize', 18);
ylim([-1.5,1.5])
grid on;
```

```
grid minor;

% Phase spectrum

subplot(2,1,2)
stem(k, angle(Xk), 'b', 'LineWidth',2);
hold on;
xlabel('Time (s)');
title('Phase Spectrum','Interpreter', 'latex');
%legend('Square Wave', 'Fourier Series Approximation');
set(gca, 'XColor', [0, 0, 0], 'YColor', [0, 0, 0], 'TickDir', 'out');
xaxis = get(gca, 'XAxis');
xaxis.TickLabelInterpreter = 'latex';
yaxis = get(gca, 'YAxis');
yaxis.TickLabelInterpreter = 'latex';
xlabel('k','Interpreter', 'latex');
ylabel('$\angle X_k$','Interpreter', 'latex');
set(gca, 'FontSize', 18);
ylim([-2.5,2.5])
grid on;
grid minor;

exportgraphics(f, '../figures/CPE381_FA24_HW04_Q5_Spectrum.pdf');

function result = X_k(k)
    Amplitude = 3.0;
    result = zeros(size(k));

    % Handle k == 0 separately
    idx = (k == 0);
    result(idx) = Amplitude/2;

    % Calculate for non-zero k
    idx = (k ~= 0);
    result(idx) = (Amplitude/2) .* (1 - (-1).^k(idx)) ./ (1j .* k(idx) .* pi);
end
```

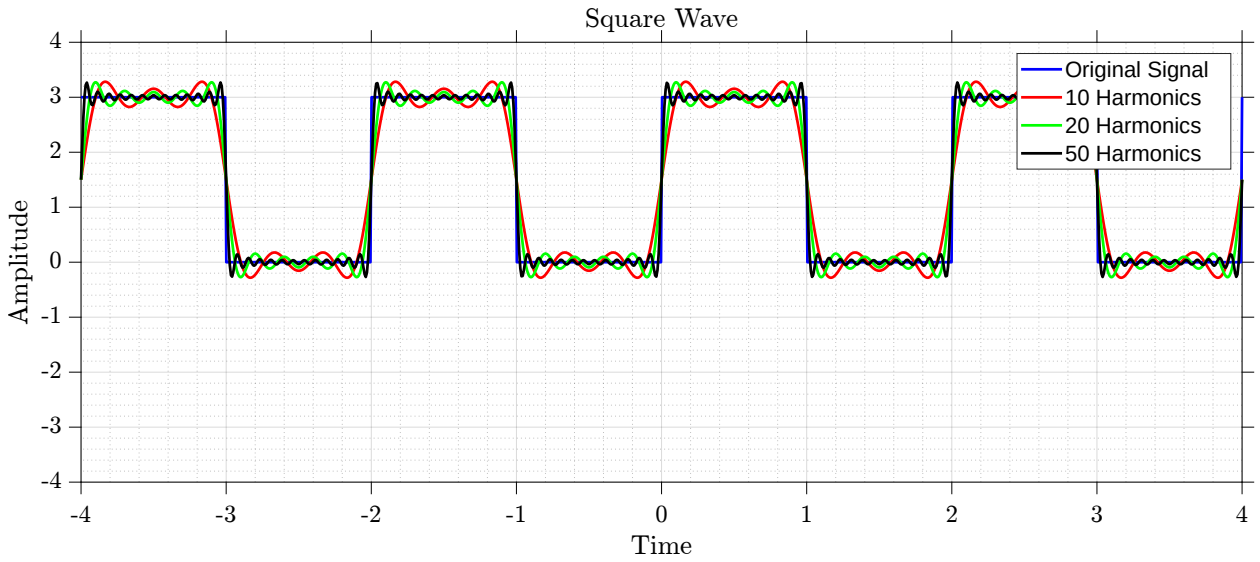


Figure 3: Harmonics

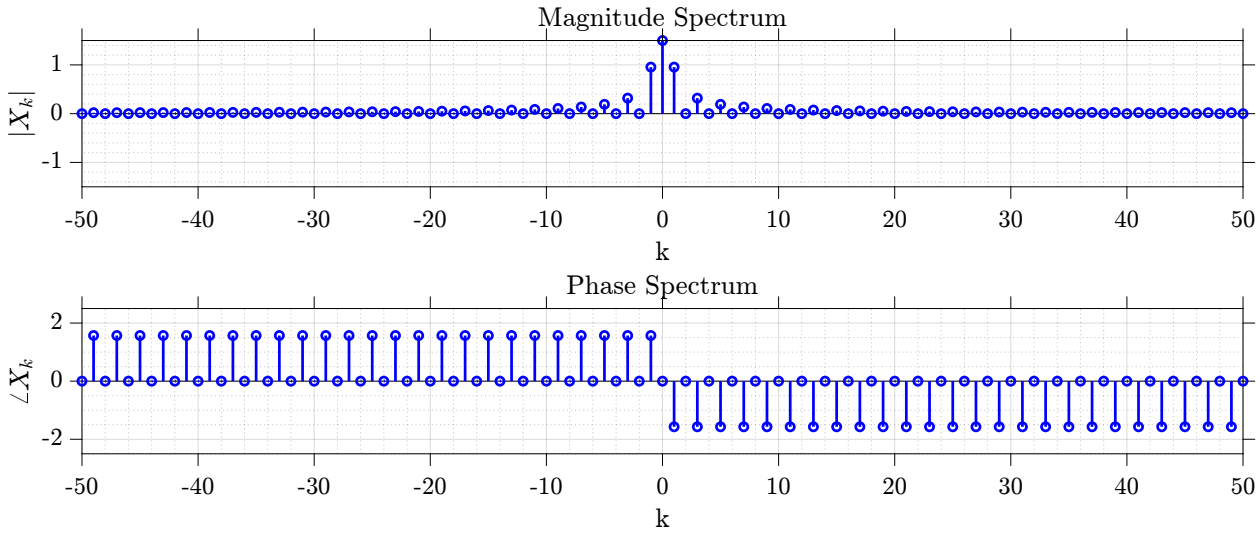


Figure 4: Harmonics

We observe that as you add more and more harmonics, the signal approximates to the true square waves. However, you will notice that there will be some kind of ripple effect there.

6 Signal Smoothing Using Fourier Series (30 Points)

Part 1 (5 Points)

Consider the MATLAB code provided in https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/Ch04_Synthetic_Power_Data_With_Distorted_Harmonics.m.

In this code, we see that odd number of harmonics (3, 5, 7, 11, 13) with small percentage of amplitude is added to the original voltage signal to create distorted harmonics along with some random noise.

Modify the code to add 15th harmonic with 8% amplitude distortion (i.e. 0.08) to the original voltage signal. Compute reconstruction error (opposite of the reconstruction quality) when approximating the noisy voltage using 12 harmonics. Report the answer. Also report the total harmonic distortion THD value.

Please include the code with the same file as originally provided in the course GitHub codebase. **(5 Points)**

Part 2

Download a datset provided in https://github.com/rahulbhadani/CPE381_FA26/blob/main/Data/main_power_subset.csv. The dataset contain time and amplitude column.

For those who are interested in the background, the data was obtained from Oak Ridge National Lab's repository (https://doi.ccs.ornl.gov/dataset/13e3dc9a-b06f-5752-93cf-f25a8669d87f). Only a subset of this large dataset has been used for this part of the assignment.

The given CSV file contains apparent voltage value from a 480 Volt power line enterin Oak Ridge National Lab's one of the buildings measured with a Rogowski coil.

You are required to submit a MATLAB script with the exact name `harmonics_oak_ridge.m` where you will read the CSV file as a table and perform the following task using the data provided in the file:

1. Plot the amplitude from the csv file as a function of time. Include the necessary code and the generated plot as a part of your submission. **(2 Points)**
2. Consider that the fundamental frequency of the signal is $60\ Hz$ as is the case with the power grid in the United States, compute the trigonometric fourier coefficients a_n corresponding to cosine part of the trigonometric Fourier series, and b_n corresponding to the sine part of the trigonometric Fourier series. Report the DC term separately. **(9 +1 Points)**
3. Report the total harmonic distortion based on the formula provided in the code mentioned in Part 1. **(2 Points)**
4. Create a bar plot for harmonic amplitude $A = \sqrt{\frac{1}{2}(a_k^2 + b_k^2)}$, as a function of harmonic number k. Harmonic amplitude is the root mean square of $|X_k|$ provided in Page 255 of the textbook. Include the bar plot figure as a part of your submission. **(5 Points)**

5. Next, reconstruct the signal by using only 3 harmonics, and then 5, 15, and 20 harmonics. Then create four plot showing original signal overlaid with reconstructed signal for all four cases. Choose colors appropriately for clarity. Include all the figures in your submission. Compute the root mean square (RMS) error for each four cases when compared with the original signal and report them as a part of your submission. Report all RMS error as a part of your submission. **(6 Points)**

7 Answer:

Output:

Code available at https://github.com/rahulbhadani/CPE381_FA25/blob/main/Code/HW04_distortion.m

Total Harmonic Distortion (THD): 44.30%

DC Power: 0.0026 W

Reconstruction Quality (12 harmonics): 90.65%

Part 2

Download a datset provided in https://github.com/rahulbhadani/CPE381_FA25/blob/main/Data/main_power_subset.csv. The dataset contain time and amplitude column.

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Answers:

Code also at https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/HW04_main_power_subset.m

```
% MATLAB Code to smooth main power data using Fourier Series
% Flat code without functions or classes

clear; clc;

% ========== LOAD DATA FROM CSV ==========
fprintf('Loading main power subset data...\n');
data_table = readtable('../Data/main_power_subset.csv');
time = data_table.time_seconds;
signal = data_table.amplitude_raw;

fprintf('Data loaded: %d points\n', length(signal));
fprintf('Duration: %.4f seconds\n', time(end) - time(1));

% ========== PARAMETERS ==========
T = time(end) - time(1);  % Period (total duration)
fs = length(time) / T;     % Sampling frequency
fundamental_freq = 60;     % Hz (power grid fundamental)
max_harmonics = 20;        % Number of harmonics to compute

fprintf('\nSampling rate: %.0f Hz\n', fs);
fprintf('Fundamental frequency: %.0f Hz\n', fundamental_freq);
fprintf('Max harmonics: %d\n', max_harmonics);

% ========== COMPUTE FOURIER COEFFICIENTS ==========
fprintf('\nComputing Fourier coefficients...\n');

omega0 = 2 * pi * fundamental_freq;

% DC component (a0)
a0 = 2 * trapz(time, signal) / T;
```

```
% Initialize coefficient arrays
an_coeffs = zeros(1, max_harmonics);
bn_coeffs = zeros(1, max_harmonics);

% Calculate harmonics using numerical integration (trapezoidal rule)
for n = 1:max_harmonics
    % Cosine coefficients
    integrand_cos = signal .* cos(n * omega0 * time);
    an_coeffs(n) = 2 * trapz(time, integrand_cos) / T;

    % Sine coefficients
    integrand_sin = signal .* sin(n * omega0 * time);
    bn_coeffs(n) = 2 * trapz(time, integrand_sin) / T;

    if mod(n, 5) == 0
        fprintf('  Computed harmonic %d/%d\n', n, max_harmonics);
    end
end

fprintf('Fourier coefficients computed!\n');
fprintf('DC component (a0): %.4f\n', a0);

% ========== CALCULATE HARMONIC AMPLITUDES ==========
harmonic_amplitudes = zeros(1, max_harmonics);
for n = 1:max_harmonics
    harmonic_amplitudes(n) = sqrt(0.5 * (an_coeffs(n)^2 + bn_coeffs(n)^2));
end

% ========== CALCULATE THD ==========
fundamental_power = 0.5 * (an_coeffs(1)^2 + bn_coeffs(1)^2);
harmonic_power = 0;
for n = 2:max_harmonics
    harmonic_power = harmonic_power + 0.5 * (an_coeffs(n)^2 + bn_coeffs(n)^2);
end

if fundamental_power > 0
    thd = sqrt(harmonic_power / fundamental_power) * 100;
else
    thd = 0;
end

fprintf('\nTotal Harmonic Distortion (THD): %.2f%%\n', thd);

% ========== RECONSTRUCT SIGNAL WITH DIFFERENT HARMONICS ==========
```

```
fprintf('\nReconstructing signals...\n');

% Number of harmonics for different reconstructions
num_harmonics_list = [1, 3, 5, 10, 15, 20];
reconstructed_signals = cell(1, length(num_harmonics_list));

for idx = 1:length(num_harmonics_list)
    num_harmonics = num_harmonics_list(idx);

    % Start with DC component
    reconstructed = (a0 / 2) * ones(size(time));

    % Add harmonics
    for n = 1:min(num_harmonics, max_harmonics)
        reconstructed = reconstructed + ...
            an_coeffs(n) * cos(n * omega0 * time) + ...
            bn_coeffs(n) * sin(n * omega0 * time);
    end

    reconstructed_signals{idx} = reconstructed;
    fprintf('  Reconstructed with %d harmonics\n', num_harmonics);
end

% ========== PLOTTING ==========
fprintf('\nCreating plots...\n');

f  = figure('Position', [100, 100, 1400, 1000]);

% Plot 1: Original signal (time domain)
subplot(3, 3, 1);
plot(time * 1000, signal, 'b-', 'LineWidth', 1);
title('Original Signal (Full)');
xlabel('Time (ms)');
ylabel('Amplitude (raw)');
grid on;

% Plot 2: Original signal (zoomed - first 5 cycles)
subplot(3, 3, 2);
cycles_to_show = 5;
duration_cycles = cycles_to_show / fundamental_freq;
idx_zoom = time <= duration_cycles;
plot(time(idx_zoom) * 1000, signal(idx_zoom), 'b-', 'LineWidth', 1.5);
title(sprintf('Original Signal (First %d cycles)', cycles_to_show));
xlabel('Time (ms)');
```

```
ylabel('Amplitude (raw)');
grid on;

% Plot 3: Harmonic spectrum
subplot(3, 3, 3);
bar(1:max_harmonics, harmonic_amplitudes);
title('Harmonic Spectrum');
xlabel('Harmonic Number');
ylabel('Amplitude');
grid on;
text(max_harmonics*0.6, max(harmonic_amplitudes)*0.9, sprintf('THD: %.2f%%', thd), ...
    'FontSize', 10, 'BackgroundColor', 'white');

% Plot 4-9: Reconstructed signals with different numbers of harmonics
for idx = 1:length(num_harmonics_list)
    subplot(3, 3, 3 + idx);
    num_harmonics = num_harmonics_list(idx);
    reconstructed = reconstructed_signals{idx};

    % Plot zoomed version for better visibility
    plot(time(idx_zoom) * 1000, signal(idx_zoom), 'b-', 'LineWidth', 1, 'DisplayName', 'Original'
    hold on;
    plot(time(idx_zoom) * 1000, reconstructed(idx_zoom), 'r--', 'LineWidth', 1.5, ...
        'DisplayName', sprintf('%d harmonics', num_harmonics));
    hold off;

    title(sprintf('Reconstructed (%d harmonics)', num_harmonics));
    xlabel('Time (ms)');
    ylabel('Amplitude');
    legend('show', 'Location', 'best');
    grid on;

    % Calculate reconstruction error
    error_rms = sqrt(mean((signal - reconstructed).^2));
    fprintf("RMS Error for %d harmonics are: %f\n", num_harmonics, error_rms);
    error_percent = (error_rms / std(signal)) * 100;
    fprintf("RMS ERROR PERCENT for %d harmonics are: %f\n", num_harmonics, error_percent);

    text(min(time(idx_zoom))*1000 + 1, max(signal(idx_zoom))*0.9, ...
        sprintf('Error: %.2f%%', error_percent), 'FontSize', 8, 'BackgroundColor', 'white');
end

sgtitle('Fourier Series Analysis: Main Power Signal Smoothing', 'FontSize', 14, 'FontWeight', 'bo
```

```
exportgraphics(f, 'figures/HW04_main_power_subset_analysis.pdf');


% ========== SAVE SMOOTHED DATA ==========
fprintf('\nSaving smoothed data...\n');

% Choose the best reconstruction (e.g., 10 harmonics for good smoothing)
smoothed_signal = reconstructed_signals{4};  % 10 harmonics

% Create output table
output_table = table(time, signal, smoothed_signal, ...
    'VariableNames', {'time_seconds', 'original_signal', 'smoothed_signal'});

% Save to CSV
writetable(output_table, 'main_power_smoothed.csv');
fprintf('Smoothed data saved to: main_power_smoothed.csv\n');


fprintf('Original signal: %d points\n', length(signal));
fprintf('DC component: %.4f\n', a0);
fprintf('Fundamental amplitude: %.4f\n', harmonic_amplitudes(1));
fprintf('THD: %.2f%%\n', thd);
fprintf('Recommended harmonics for smoothing: 10-15\n');
fprintf('Top 3 harmonic amplitudes:\n');
[sorted_amps, sorted_idx] = sort(harmonic_amplitudes, 'descend');
for i = 1:min(3, length(sorted_amps))
    fprintf('  Harmonic %d: %.4f\n', sorted_idx(i), sorted_amps(i));
end
```

Output:

DC Component: -257.0216

Total Harmonic Distortion: 79.65%

RMS Error for 1 harmonics are: 3325.032701

RMS ERROR PERCENT for 1 harmonics are: 80.413093

RMS Error for 3 harmonics are: 3329.747677

RMS ERROR PERCENT for 3 harmonics are: 80.527120

RMS Error for 5 harmonics are: 3000.174184

RMS ERROR PERCENT for 5 harmonics are: 72.556665

RMS Error for 10 harmonics are: 2921.359331

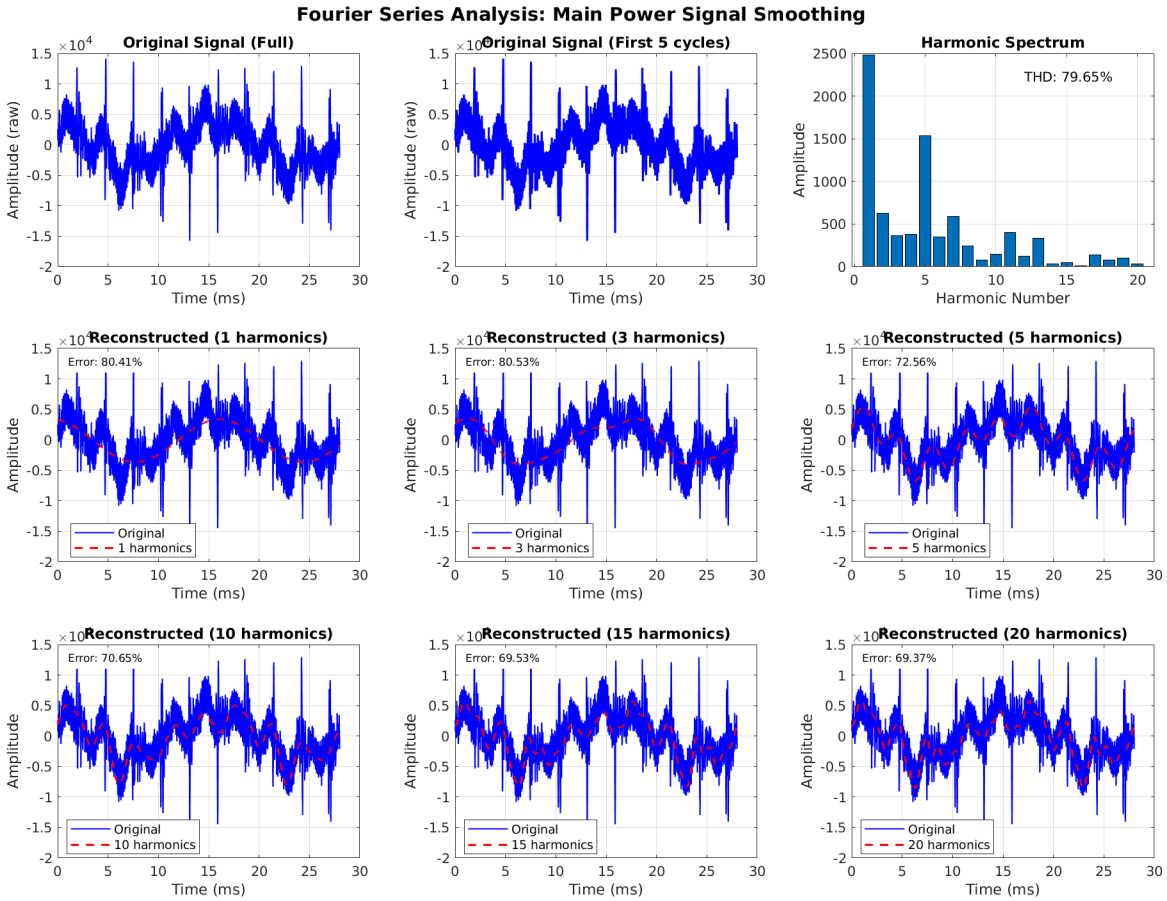
RMS ERROR PERCENT for 10 harmonics are: 70.650595

RMS Error for 15 harmonics are: 2875.055272

RMS ERROR PERCENT for 15 harmonics are: 69.530771

RMS Error for 20 harmonics are: 2868.448363

RMS ERROR PERCENT for 20 harmonics are: 69.370988



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